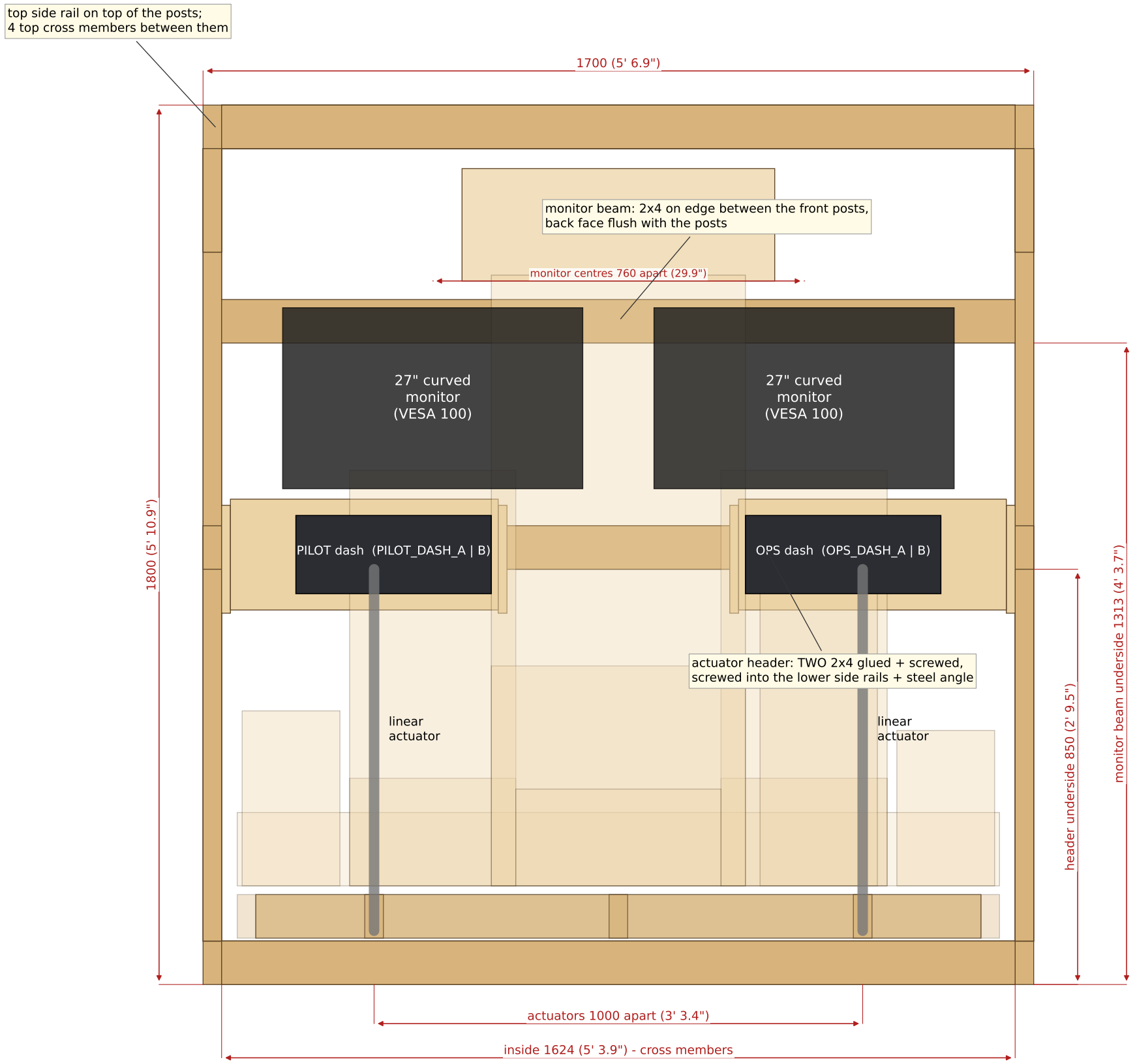
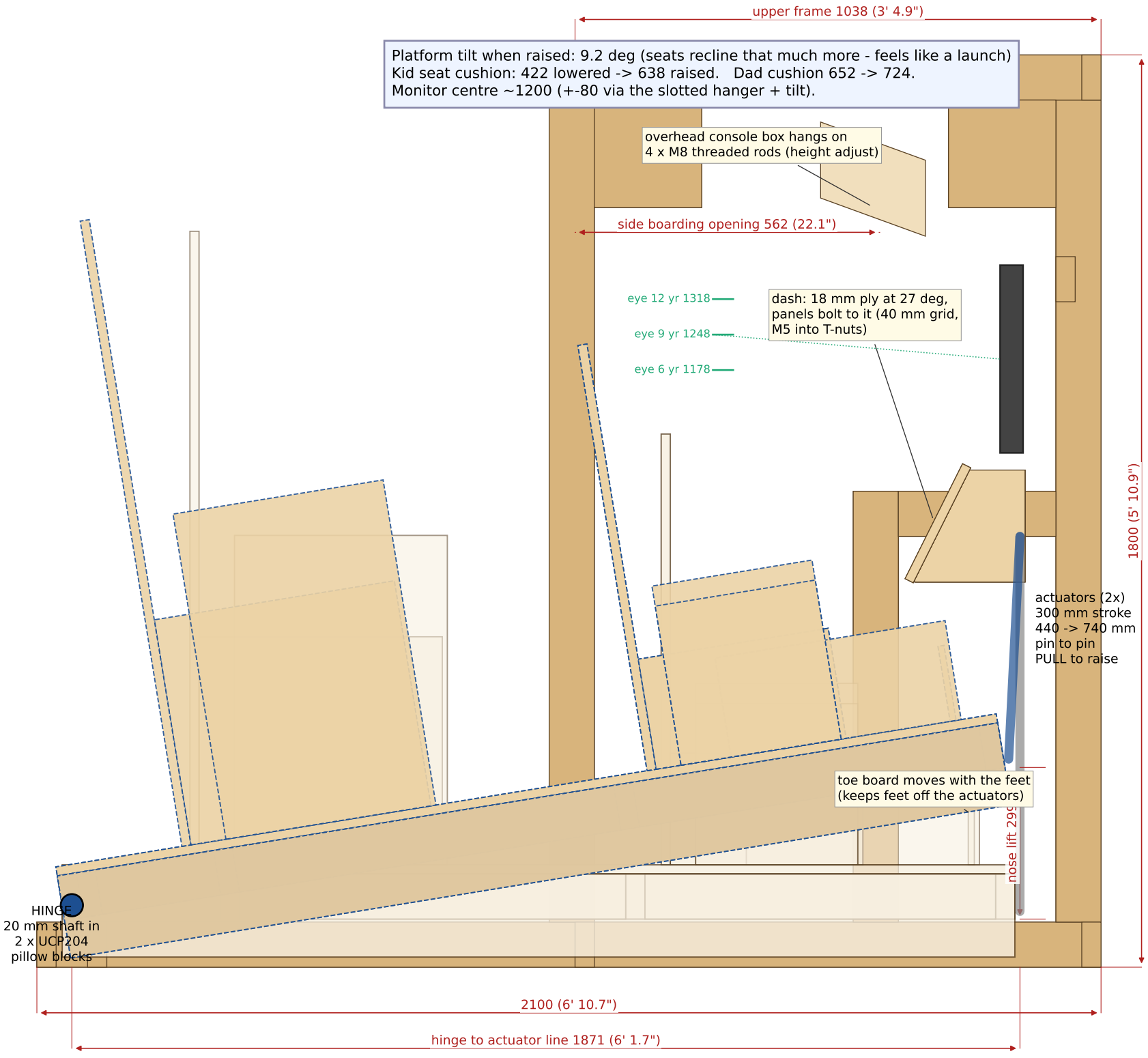
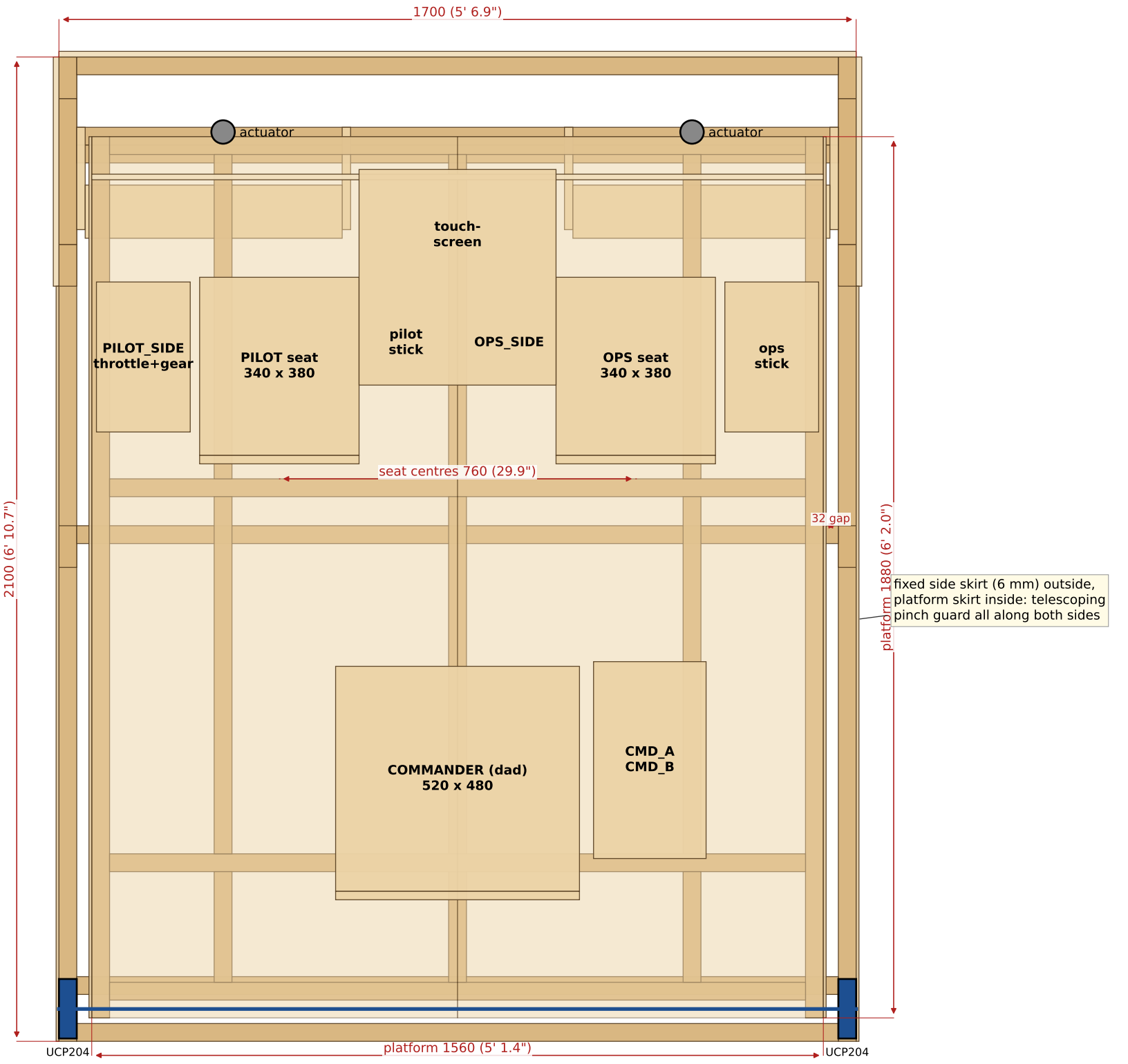
1. Front elevation (looking at the front of the cockpit from outside; skins removed)



2. Side elevation (from the RIGHT, front of cockpit at the right) - platform LOWERED solid, RAISED dashed



3. Plan view (from above, front at the top) - top frame omitted for clarity



4. Seat-lift geometry, actuator sizing and hardwired control circuit

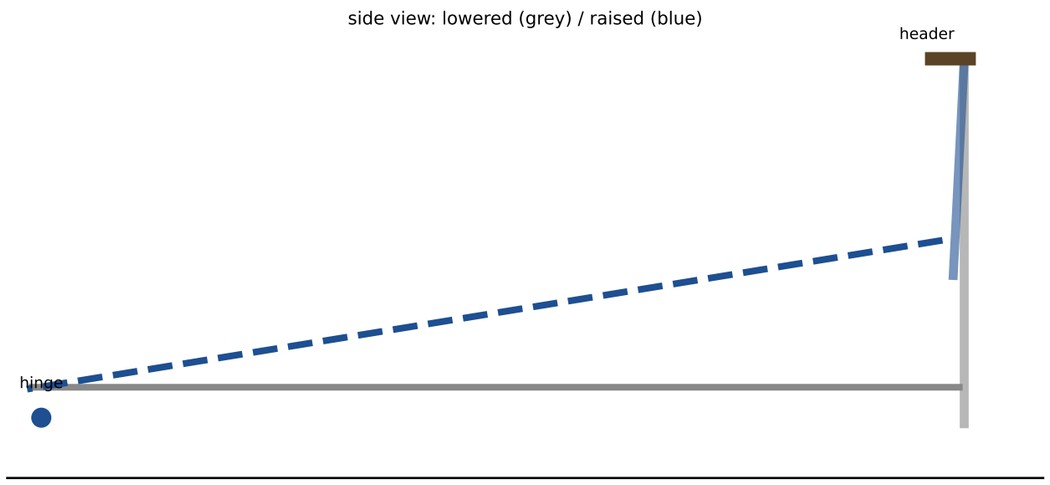
```
ASSUMPTIONS (change in frame/frame_model.py and re-run):
  - two kids (2 x 45 kg, upper bound for 12 yr): 90 kg at 660 mm behind the front (= 1371 mm in front of the hinge)
  - adult (dad): 110 kg at 1520 mm behind the front (= 511 mm in front of the hinge)
  - platform, deck, seats, consoles (estimate): 110 kg at 1150 mm behind the front (= 881 mm in front of the hinge)
  - hinge at 2031 mm behind the front, 122 mm above the floor; actuators 1871 mm in front of the hinge
  - actuators hang from the header (top clevis z=850) and PULL the platform nose up (bottom clevis z=110 lowered)
  - stroke 300 mm, retracted pin-to-pin 440 mm -> raised angle 9.2 deg

RESULT (both actuators together / each):
  angle  0.0 deg: moment   2713 N*m, lever arm 1871 mm, pull  1450 N total =  725 N (163 lbf) each, length 740 mm
  angle  4.6 deg: moment   2611 N*m, lever arm 1865 mm, pull  1400 N total =  700 N (157 lbf) each, length 589 mm
  angle  9.2 deg: moment   2492 N*m, lever arm 1832 mm, pull  1360 N total =  680 N (153 lbf) each, length 440 mm

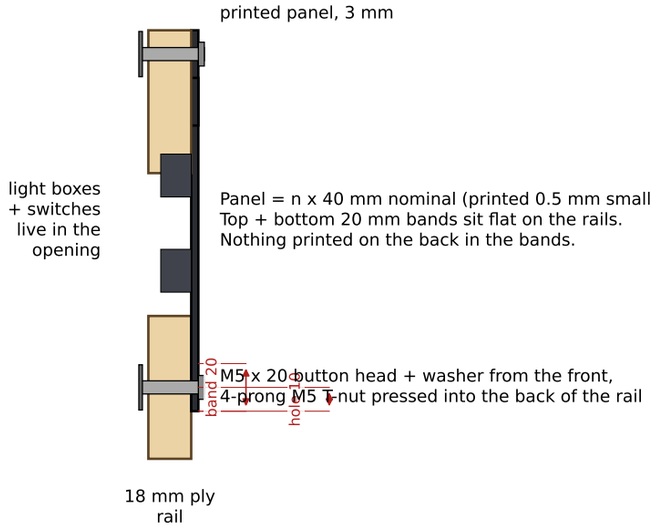
BUY: 2 x 12 V linear actuators, 300 mm (12") stroke, >= 1500 N (330 lbf+) rated in PULL as well as push,
     built-in limit switches, ~10-15 mm/s (slow is safer: about 25 s for full travel), clevis ends + steel mounting brackets
     (the brackets that come with the actuators - do NOT print these). Safety factor >= 2 on the worst case above.
     Same model for both, wired in parallel; the stiff platform keeps them together. Expect 3-5 A each under load:
     12 V 20 A supply, 15 A blade fuse, 14 AWG (2 mm^2) wire.

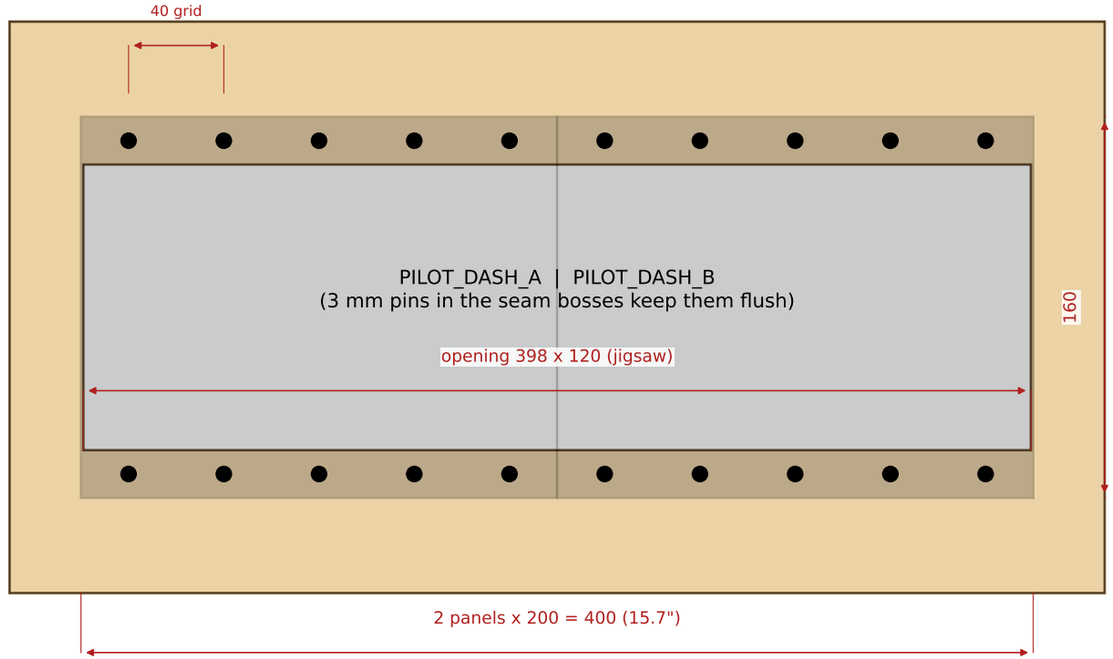
HARDWIRED CONTROL (no microcontroller can move the seats):
  12V+ --[15A fuse]--[KEY SWITCH 'SEAT KEY']--[E-STOP (NC, latching)]--+--> DPDT (ON)-OFF-(ON) 'SEAT LIFT' (hold-to-run, spring return)
                                                                      |
  UP position  : actuator +/- polarity A --[UP limit switch, NC]---- actuators (in parallel)
  DOWN position: polarity B ----------------[DOWN limit switch, NC]-- actuators
  Each external limit switch has a diode across it (1N5408, 3 A) so the switch only blocks motion PAST its end and
  you can always drive back the other way. Actuators' own internal limit switches are a second layer.
  A separate dry contact (2nd micro switch or the limit switch's NO pole) goes to Falcon Command so the game can
  READ 'seats up' - it never drives anything.  If the 12 mm toggle is not rated for the motor current, let it switch
  two 30 A automotive relays (still hardwired, still hold-to-run).

PINCH-POINT RULES used in this frame (ISO 13854 minimum gaps: finger 25, hand 100, arm 120, foot 120, leg 180 mm):
  - platform to posts: 32 mm side gap + telescoping 6 mm skirts (fixed outside, moving inside) along both sides
  - feet: toe board fixed to the platform in front of the kids' feet (moves with them) hides the actuators;
    toe-board top to dash cheek gap at full lift >= 125 mm
  - front + side skins enclose the actuator bay; no hands can reach the actuator rods
  - dad holds the switch and watches; letting go stops everything; E-stop within reach; KEY removed = no motion
  - rule for the kids: hands on lap, harness on, 'SEATS MOVING' callout before every move
```



5. Panel bay detail - how every printed panel bolts to the plywood (40 mm grid, M5 into T-nuts)





Drill the T-nut holes with a printed drilling template: print any panel's
single-material body at 15% infill as a jig, or mark through the panel's own M5 holes.
T-nut hole: 7 mm (check your T-nuts), from the back, tap the T-nut home with a hammer.